

Subspaces

(1)

Defⁿ of subspace: Let V be a vector space over a field IK and W be a subset of V . Then W is a subspace of V if W is itself a vector space over IK with respect to the operations of vector addition and scalar multiplication on V .

(First Fundamental Theorem of subspace)
Prove that a subset W of a vector space V is called a subspace of V

iff

(a) W is non empty

(b) W is closed under vector addition

$$\forall u, v \in W \Rightarrow u+v \in W \text{ and}$$

(c) W is closed under scalar multiplication, $\forall k \in IK$ and

$$\forall v \in W \Rightarrow kv \in W.$$

Proof: Suppose W satisfies (a), (b) and (c).
Now by (a) W is non-empty and by (b) and (c) the operations of vector addition and scalar multiplication are well defined in W .

Again, the vectors in W belongs to V ,
then the following axioms hold in W .

$$(i) \quad u+v = v+u$$

$$(ii) \quad (u+v)+w = u+(v+w)$$

$$(iii) \quad k(u+v) = ku + kv, \quad \forall u, v, w \in W \\ \text{and } k, k' \in K.$$

$$(iv) \quad (k+k')u = ku + k'u$$

$$(v) \quad (kk')u = k(k'u)$$

$$(vi) \quad \text{For the unit scalar } 1 \in K, 1u = u, \\ \forall u, v, w \in W.$$

Now we only need to show that

(A) Existence of zero vector $0 \in W \Rightarrow$
 $u+0 = u \quad \forall u \in W.$

(B) Existence of additive inverse:
 $\forall u \in W, \exists -u \in W \Rightarrow u+(-u) = 0$

Now we will prove (A).

By (a) W is non empty say $u \in W$,

then by (c) $0u = 0 \in W$ and $u+0 = u \quad \forall u \in W$

Hence W satisfies (A)

Now we will prove (B).

By (a), if $u \in W$

Then by (c) $(-1)u = -u \in W$

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$$\text{and } u + (-u) = 0 \quad \forall u \in W$$

Hence W satisfies (B)

therefore all the properties are satisfied by W .

So, W is a subspace of vector space V .

Conversely, if W is a subspace of V , then clearly (a), (b), (c) hold in W .

$$(i) 0 \in W \text{ i.e. } W \neq \emptyset$$

$$(ii) \forall u, v \in W \Rightarrow au + bv \in W, \forall a, b \in K$$

Since they are the axioms of the vector space.

2nd Fundamental Theorem of Subspace:

Theorem!

Let W be a subset of a vector space $V(F)$. Then show that W is a subspace of $V(F)$ iff

$$(i) 0 \in W \text{ i.e. } W \neq \emptyset \text{ and}$$

$$(ii) \forall u, v \in W \Rightarrow au + bv \in W,$$

$$\forall a, b, c \in F$$

Proof: Suppose W satisfies (i) and (ii)

then by (i) $0 \in W \Rightarrow W$ is non-empty

Again if $u, v \in W$ then by (ii) $u+v = 1u+1v$
 $\in W$

Since it is a vector space of the form

$$au+bv.$$

Thus W is closed under vector addition.

Again, if $v \in W$ and if $k \in F$ then by (ii)

we have, $k v = k v + 0 = k v + 0 v \in W$

Since for $0 \in F$ and $\forall v \in W$, $0 v = 0$

Therefore, W is closed under scalar multiplication.

Thus, W satisfies the three conditions of the fundamental theorem and therefore W is a subspace of $V(F)$.

conversely, if W is a subspace of the vector space $V(F)$.

then clearly conditions (i) and (ii) hold for W .

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Theorem! Let W_1 and W_2 be two subspaces of a vector space $V(F)$. Then show that $W_1 \cap W_2$ is a subspace of $V(F)$.

proof: Since W_1 and W_2 are subspaces of W_1 and $0 \in W_2$

Thus $0 \in W_1 \cap W_2$

$$\Rightarrow W_1 \cap W_2 \neq \emptyset$$

Again, suppose $u, v \in W_1 \cap W_2$. Then

$$\left. \begin{array}{l} u, v \in W_1 \Rightarrow au + bv \in W_1 \\ u, v \in W_2 \Rightarrow au + bv \in W_2 \end{array} \right\} \forall a, b \in F$$

Since W_1 and W_2 are subspaces.

Thus $\forall a, b \in F$ and $\forall u, v \in W_1 \cap W_2$

$$\Rightarrow au + bv \in W_1 \cap W_2$$

Thus $W_1 \cap W_2$ is a subspace of $V(F)$.

Determine whether or not
 $W = \{ (a, b, c) : a^2 + b^2 + c^2 \geq 1 \}$ is a subspace of $\mathbb{R}^3$.

Soln: Let $0 = (0, 0, 0)$
Since $0^2 + 0^2 + 0^2 = 0 \not\geq 1$

$\therefore 0 \notin W$.

Hence W is not a subspace of $\mathbb{R}^3$.

Determine whether or not
 $W = \{ (a, b, c) : a + 2b + 3c = 0 \}$ is a subspace of $\mathbb{R}^3$.

Soln: Let $0 = (0, 0, 0) \in W$

Since $0 + 2 \cdot 0 + 3 \cdot 0 = 0$ and $W \neq \emptyset$

$\therefore W$ is non-empty.

Let $u = (a_1, b_1, c_1)$ and $v = (a_2, b_2, c_2) \in W$

then, $a_1 + 2b_1 + 3c_1 = 0$

$a_2 + 2b_2 + 3c_2 = 0$

Now, for any $k_1, k_2 \in \mathbb{R}$ we have,

$$k_1 u + k_2 v = k_1 (a_1, b_1, c_1) + k_2 (a_2, b_2, c_2)$$

$$= (k_1 a_1 + k_2 a_2, k_1 b_1 + k_2 b_2, k_1 c_1 + k_2 c_2)$$

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$$\begin{aligned}
 \text{Now, } k_1 a_1 + k_2 a_2 + 2(k_1 b_1 + k_2 b_2) + 3(k_1 c_1 + k_2 c_2) \\
 &= k_1 \left(\underbrace{a_1 + 2b_1 + 3c_1}_0 \right) + k_2 \left(\underbrace{a_2 + 2b_2 + 3c_2}_0 \right) \\
 &= k_1 \cdot 0 + k_2 \cdot 0 \\
 &= 0
 \end{aligned}$$

Thus, $k_1 u + k_2 v \in W$, $\forall k_1, k_2 \in \mathbb{R}$
and $u, v \in W$.

Hence, W is a subspace of $\mathbb{R}^3$.

Let $V = \left\{ A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R} \right\}$
and $W = \{ A \in V : |A| = 0 \}$. Determine
whether or not W is a subspace of V .

Solⁿ: Let $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Since $\begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix} = 0$

Therefore $A \in W$ and $W \neq \emptyset$

Let $B = \begin{bmatrix} 5 & 2 \\ 0 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 8 \\ 0 & 2 \end{bmatrix}$

Then, $B, C \in W$.

Since $|B| = \begin{vmatrix} 5 & 2 \\ 0 & 0 \end{vmatrix} = 0$ and $|C| = \begin{vmatrix} 0 & 8 \\ 0 & 2 \end{vmatrix} = 0$

But $B + C = \begin{bmatrix} 5 & 2 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 8 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 0 & 2 \end{bmatrix} \notin W$

Since $|B+C| = \begin{vmatrix} 5 & 10 \\ 0 & 2 \end{vmatrix} = 10 \neq 0$

Hence W is not a subspace of V .

Generator / span: Let S is a nonempty subset of a vector space V (or $\mathbb{R}^n$) and if the set of all linear combinations of the vectors in S , $L(S)$ is a subspace of V (or $\mathbb{R}^n$) then the set S is called a generator of $L(S)$ or $L(S)$ is generated or spanned by S .

Example: Show that the vectors $u_1 = (1, 1, 1)$, $u_2 = (1, 2, 3)$ and $u_3 = (1, 5, 8)$ spans $\mathbb{R}^3$.

Proof: Now we have to show that, an arbitrary vector $v = (a, b, c) \in \mathbb{R}^3$ is a linear combination of u_1, u_2, u_3 .

Now,
$$(a, b, c) = c_1 (1, 1, 1) + c_2 (1, 2, 3) + c_3 (1, 5, 8)$$

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$$\Rightarrow (a, b, c) = (c_1 + c_2 + c_3, c_1 + 2c_2 + 5c_3, c_1 + 3c_2 + 8c_3)$$

this is equivalent to,

$$c_1 + c_2 + c_3 = a$$

$$c_1 + 2c_2 + 5c_3 = b$$

$$c_1 + 3c_2 + 8c_3 = c$$

now, the augmented matrix is,

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 1 & 2 & 5 & b \\ 1 & 3 & 8 & c \end{array} \right] \begin{array}{l} R_2' = R_2 - R_1 \\ R_3' = R_3 - R_1 \end{array}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 4 & b-a \\ 0 & 2 & 7 & c-a \end{array} \right] \begin{array}{l} R_2' = R_2 - R_1 \\ R_3' = R_3 - 2R_2 \end{array}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 4 & b-a \\ 0 & 0 & -1 & c-2b+a \end{array} \right] R_3' = (-1)R_3$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 4 & b-a \\ 0 & 0 & 1 & 2b-a-c \end{array} \right]$$

$$c_1 + c_2 + c_3 = a$$

$$c_2 + 4c_3 = b - a$$

$$c_3 = 2b - a - c$$

$$\Rightarrow c_2 + 4(2b - a - c) = b - a$$

$$\Rightarrow c_2 = b - a - 8b + 4a + 4c = 3a - 7b + 4c$$

$$\therefore c_1 + 3a - 7b + 4c + 2b - a - c = a$$

$$\Rightarrow c_1 = a - 2b + 5c - 3e$$

$$\Rightarrow c_1 = 5b - 3c - a$$

$$\text{so, } c_1 = -a + 5b - 3c, \quad c_2 = 3a - 7b + 4c,$$

$$c_3 = 2b - a - c$$

$\therefore$ The system has a solⁿ.

For example, if $a=1$, $b=1$ and $c=2$

then $c_1 = -2$, $c_2 = 4$ and $c_3 = -1$

$$\text{thus, } -2(1, 1, 1) + 4(1, 2, 3) - 1(1, 5, 8) = (1, 1, 2)$$

So, S spans $\mathbb{R}^3$.